Solution Requirements

Date	7 July 2026
Team ID	SWUID2026221619
Project Name	**A Comprehensive Measure of Well-Being**
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	User Input Handling	The system shall accept user inputs: Country (dropdown), Life Expectancy (float), Mean Years of Schooling (float), GNI per Capita (float), Internet Users (float) via an HTML form.	High
2	HDI Score Prediction	The system shall use a trained Linear Regression model (HDI.pkl) to predict the HDI score from the provided input features.	High
3	HDI Category Classification	The system shall classify the predicted HDI score into categories: Low (0.3–0.4), Medium (0.4–0.7), High (0.7–0.8), Very High (0.8–0.94).	High
4	Result Display	The system shall display the HDI score and category on the result page (resultnew.html) after prediction.	High
5	Navigation	The system shall provide navigation between Home (/), Prediction (/Prediction), and Result (/predict) pages.	Medium
6	Data Visualization	The system shall generate strip plots, heatmaps, and scatter plots during the ML training phase for EDA.	Medium
7	Model Persistence	The system shall save and load the trained model using Pickle (HDI.pkl) without retraining on each request.	High
8	Dataset Loading	The system shall load the HDI.csv dataset with 195 countries and 82 columns using Pandas.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance	The web application shall respond to prediction requests quickly.	Response time < 2 seconds

2	Accuracy	The ML model shall achieve high prediction accuracy.	R-squared Score ≥ 90% (Achieved: 93.74%)
3	Usability	The UI shall be user-friendly with clear labels and placeholders.	Usability rating ≥ 4/5
4	Reliability	The application shall handle invalid inputs gracefully.	No crashes on invalid input
5	Portability	The application shall run on Windows, Linux, and macOS.	Cross-platform compatibility
6	Maintainability	The code shall be modular, commented, and organized.	Code quality score ≥ 4/5
7	Scalability	The system shall support additional features without major refactoring.	Modular Flask route structure